

实测天体物理 I（光学与红外）作业

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第一题

取 $\lambda_{\text{eff}} = 4714\text{\AA}$ （对频率加权的平均波长）， $\Delta\lambda = 988\text{\AA}$ ，则有：

$$h\nu = \frac{hc}{\lambda_{\text{eff}}} = 4.213928051 \times 10^{-19} \text{ J}$$

$$\Delta\nu = \frac{c}{\lambda_{\text{eff}} - \frac{\Delta\lambda}{2}} - \frac{c}{\lambda_{\text{eff}} + \frac{\Delta\lambda}{2}} = 1.347703080 \times 10^{14} \text{ Hz}$$

以下计算过程中 D 为望远镜有效口径， d 为 *Seeing disc* 和艾里斑直径的较大者。
源信号：

$$\begin{aligned} S &= \frac{f_{\text{star}}}{h\nu} \cdot \frac{\pi}{4} D^2 \cdot QE \cdot t \\ &= \frac{1}{h\nu} \cdot (3631 \times 10^{-0.4m_{\text{star}}} \times \Delta\nu \times 10^{-26}) \cdot \frac{\pi}{4} D^2 \cdot QE \cdot t \\ &= C_1 t \end{aligned}$$

$$\begin{aligned} C_1 &= C_1(m_{\text{star}}, D) \\ &= \frac{1}{h\nu} \cdot (3631 \times 10^{-0.4m_{\text{star}}} \times \Delta\nu \times 10^{-26}) \cdot \frac{\pi}{4} D^2 \cdot QE \\ &= 4560298170 \times 10^{-0.4m_{\text{star}}} \times D^2 \end{aligned}$$

噪声信号：

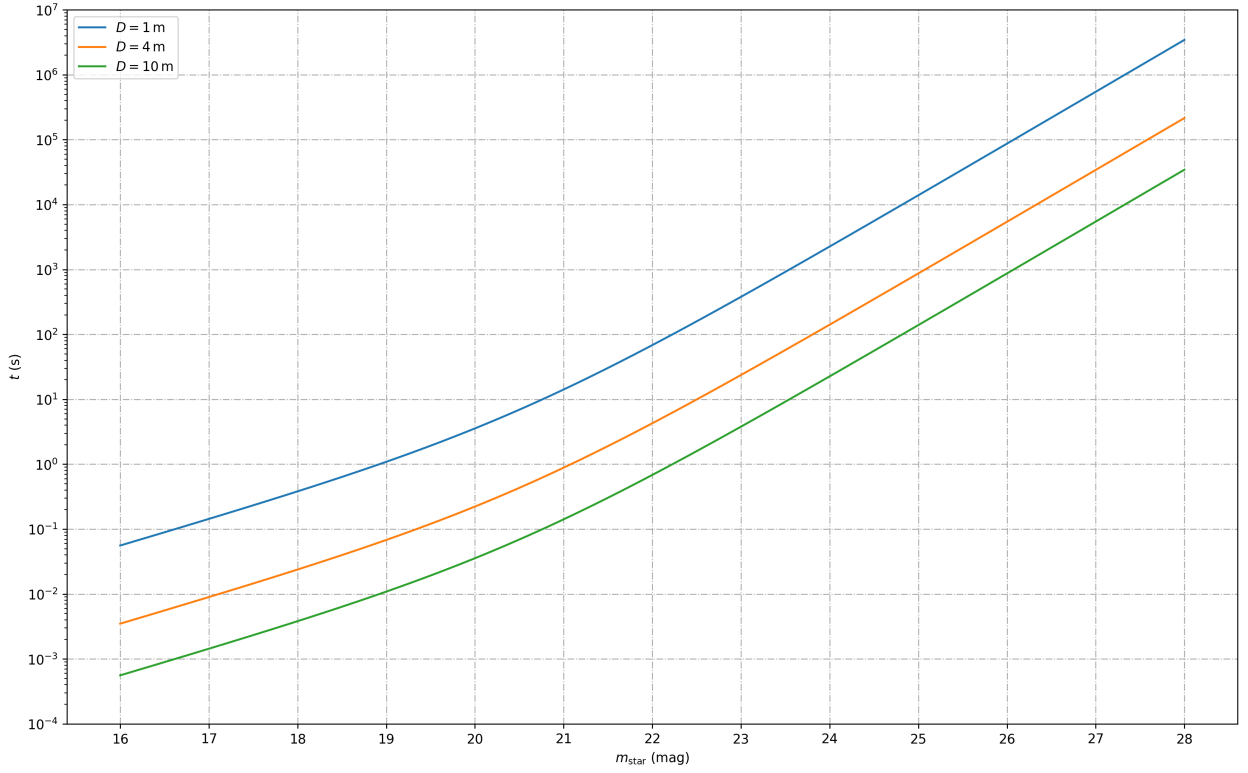
$$\begin{aligned} N &= \sqrt{\left(\sqrt{N_{\text{star}}^2 + N_{\text{sky}}^2}\right)^2 + N_{\text{sky}}^2} \\ &= \sqrt{\left(\sqrt{S + S_{\text{sky}}}\right)^2 + S_{\text{sky}}} \\ &= \sqrt{S + 2S_{\text{sky}}} \\ &= \sqrt{C_1 t + 2 \cdot \frac{f_{\text{star}}}{h\nu} \cdot \frac{\pi}{4} D^2 \cdot QE \cdot t + \frac{f_{\text{sky}}}{h\nu} \cdot \frac{\pi}{4} D^2 \cdot QE \cdot t} \end{aligned}$$

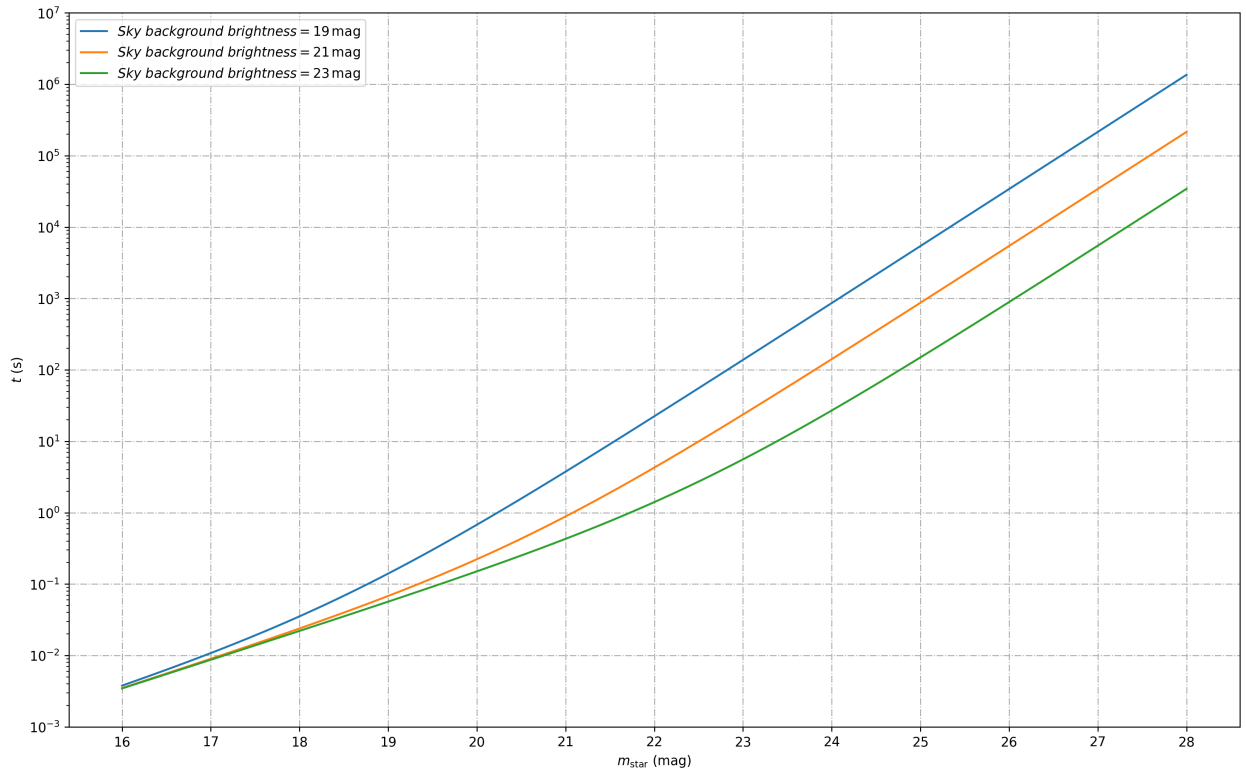
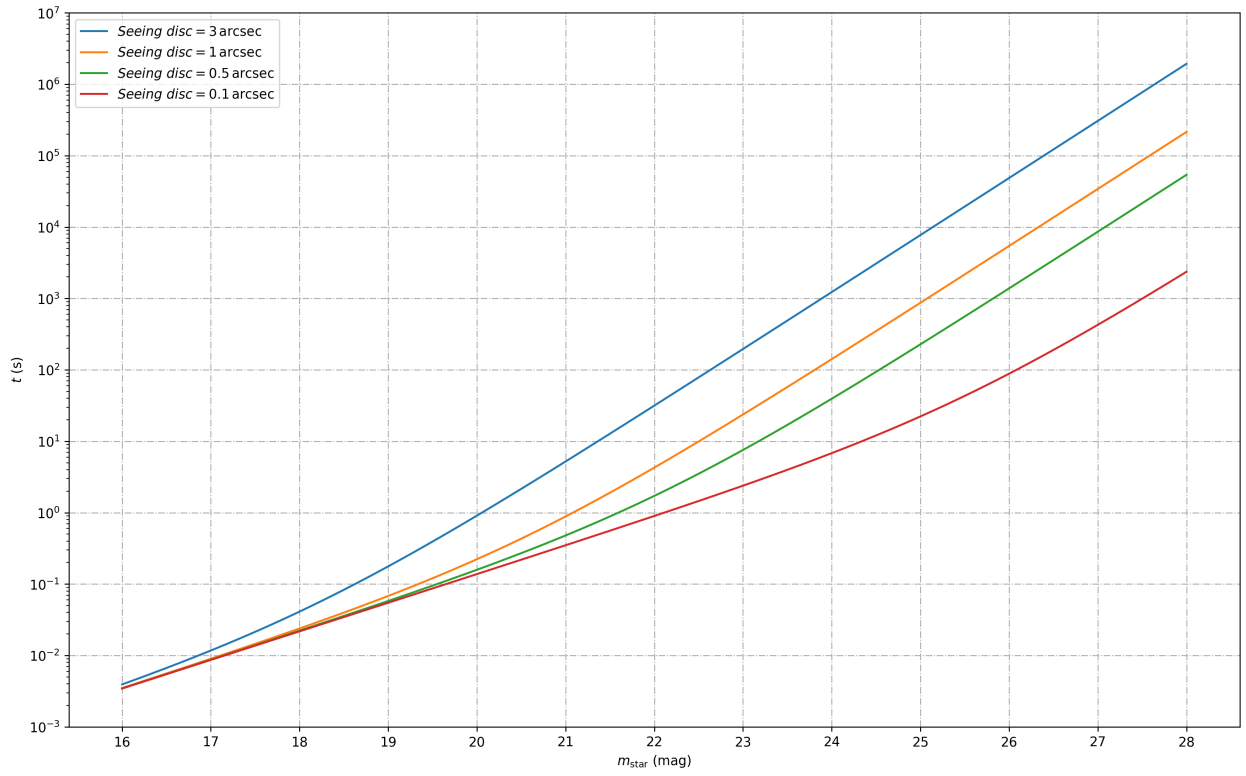
$$\begin{aligned}
&= \sqrt{C_1 t + 2 \cdot \frac{1}{h\nu} \cdot \left(3631 \times 10^{-0.4m_{\text{star}}} \times \Delta\nu \times \frac{\pi}{4} d^2 \times 10^{-26} \right) \cdot \frac{\pi}{4} D^2 \cdot QE \cdot t} \\
&= \sqrt{(C_1 + C_2)t}
\end{aligned}$$

$$\begin{aligned}
C_2 &= C_2(m_{\text{sky}}, D, d) \\
&= 2 \cdot \frac{1}{h\nu} \cdot \left(3631 \times 10^{-0.4m_{\text{sky}}} \times \Delta\nu \times \frac{\pi}{4} d^2 \times 10^{-26} \right) \cdot \frac{\pi}{4} D^2 \cdot QE \\
&= 7163299615 \times 10^{-0.4m_{\text{sky}}} \times d^2 \times D^2
\end{aligned}$$

而：

$$\begin{aligned}
\frac{S}{N} &= \frac{C_1 t}{\sqrt{(C_1 + C_2)t}} = 10 \\
t &= \frac{100(C_1 + C_2)}{C_1^2} \text{ (s)}
\end{aligned}$$





第二题

由：

$$t = \frac{100(C_1 + C_2)}{C_1^2} = 3600 \text{ s}$$

得：

$$3600 C_1^2 - 100 C_1 - 100 C_2 = 0$$

取正根：

$$C_1 = \frac{1 + \sqrt{1 + 144 C_2}}{72}$$

再由

$$C_1 = 4560298170 \times 10^{-0.4m_{\text{star}}} \times D^2$$

得：

$$m_{\text{star}} = -2.5 \log_{10} \frac{C_1}{4560298170 \times D^2}$$

注意 $D = 1 \text{ m}$ 时，艾里斑直径为 $0.11862454 \text{ arcsec} > 0.1 \text{ arcsec}$
列表如下：

$m_{\text{star}} (\text{mag})$	$m_{\text{sky}} (\text{mag/arcsec}^2)$	19	21	23
$D = 1 \text{ m}$	<i>Seeing disc</i> = 3 arcsec	22.079	23.076	24.067
	<i>Seeing disc</i> = 1 arcsec	23.267	24.257	25.231
	<i>Seeing disc</i> = 0.5 arcsec	24.013	24.993	25.942
	<i>Seeing disc</i> = 0.1 arcsec	25.532	26.446	27.236
$D = 4 \text{ m}$	<i>Seeing disc</i> = 3 arcsec	23.586	24.585	25.583
	<i>Seeing disc</i> = 1 arcsec	24.777	25.775	26.768
	<i>Seeing disc</i> = 0.5 arcsec	25.528	26.523	27.510
	<i>Seeing disc</i> = 0.1 arcsec	27.262	28.237	29.173
$D = 10 \text{ m}$	<i>Seeing disc</i> = 3 arcsec	24.581	25.581	26.580
	<i>Seeing disc</i> = 1 arcsec	25.773	26.772	27.770
	<i>Seeing disc</i> = 0.5 arcsec	26.525	27.523	28.518
	<i>Seeing disc</i> = 0.1 arcsec	28.267	29.257	30.231